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This Examination, MP491, is an Open Book examination.

Semester II Examinations 2020-21

Exam Codes	3BMS2, 3BS9, 3FM2, 4BME1, 4BMS2, 4BPT2, 4FM2
Exam	Third Science, Fourth Science Third and Fourth Financial Mathematics and Economics Third and Fourth Mathematics and Education
Module	Non-Linear Systems
Module Code	MP491
Paper No.	1
External Examiners	Professor Y. Fu
Internal Examiners	Dr Bharat B Tripathi, Dr Martin Meere
Instructions	Marks are awarded for Question 1 in Section A and the TWO best questions in Section B. All questions carry the same weight.
Duration	TWO HOURS
No. of Pages	Four pages
Discipline	Applied Mathematics
Requirements	Release to Library: Yes
MCQ	No

Section A - Answer all parts (a)-(d). ([5 Marks]/part)

1. (a) Consider the nonlinear ordinary differential equation

$$\frac{dx}{dt} = x(x^2 - 4)$$

Find all equilibrium points and determine their stability.

- (b) Consider the linear system of ordinary differential equations

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Classify the equilibrium point $(x, y) = (0, 0)$ and sketch the phase-plane portrait.

- (c) Consider the system of nonlinear ordinary differential equations given by

$$\begin{aligned} \frac{dx}{dt} &= 2x + 3y + 2x^2 - 6xy, \\ \frac{dy}{dt} &= x - 2y + 3y^2 - 4xy. \end{aligned}$$

- i. Show that this system is Hamiltonian.
 - ii. Find the Hamiltonian function.
- (d) Consider the map

$$x_{n+1} = x_n(x_n - 2)$$

for $n = 0, 1, 2, \dots$. Find all fixed points and determine if they are stable or unstable.

Section B - Answer any two questions.

2. (a) Consider the one-parameter family of nonlinear ordinary differential equations

$$\frac{dx}{dt} = x^3 - (a - 3)x,$$

where a is a real parameter.

- i. Find all equilibrium points.
- ii. Sketch their phase-portrait indicating their stability with variation in a .
- iii. Sketch the bifurcation diagram and find out the bifurcation point and the type of bifurcation.

[10 Marks]

- (b) Consider the linear 2D system of ordinary differential equations

$$\begin{aligned}\frac{dx}{dt} &= x + (3 - a)y, \\ \frac{dy}{dt} &= x + y,\end{aligned}$$

where a is a real parameter.

Investigate the nature of the equilibrium point $(0, 0)$ as the parameter a varies.

[10 Marks]

3. Consider the system of ordinary differential equations

$$\begin{aligned}\frac{dx}{dt} &= y - x^2 + 1, \\ \frac{dy}{dt} &= x^2 + y - 1.\end{aligned}$$

- (a) Locate and classify all equilibrium points of the system.

[6 Marks]

- (b) Identify, and sketch separately, the isoclines corresponding to $k = -1, 0, 1, \infty$, where

$$\frac{dy}{dx} = k.$$

[10 Marks]

- (c) Use all four isoclines found in (b) to illustrate the phase-plane portrait for this nonlinear system.

[4 Marks]

4. Consider the one-dimensional map

$$x_{n+1} = \frac{ax_n}{1 - x_n}, \quad x_n \neq 1$$

where $n = 0, 1, 2, \dots$ and a is a parameter.

- (a) Find all fixed points and investigate for what values of a the model only has non-negative fixed points.

[4 Marks]

- (b) Calculate the stability of the fixed points found in (a) when varying $a \in (-\infty, \infty)$.

[7 Marks]

- (c) Sketch the bifurcation diagram for $-2 \leq a \leq 2$.

[5 Marks]

- (d) Use the results in (a)-(c) to determine if x can undergo any bifurcations when $x > 0$. If yes, then locate all the of the bifurcation points.

[4 Marks]